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Entity Framework Core: An Introduction

Entity Framework (EF 5.0) Core is a modern object-relational mapper (ORM) or data layer class for .NET. It supports Language Integrated Query (LINQ), change tracking, updates, schema migrations, etc. It's an open source, lightweight, extensible, cross-platform version of the Entity Framework which is used only on Windows. With the release of EF Core, one can create a database mapper class that runs on any operating system (Windows, Linux and Mac).

Before we take a look at Entity Framework (EF) Core, one should be aware of the unification of the .NET Framework to .NET Core and then to .NET 5. Yes, you heard it right. From now on, .NET Core will get more updates because of cross-platform support. The .NET Framework, which is compatible with Windows only, will continue to exist but it is recommended to convert it to .NET 5 (.NET Core version). For more details, do read my article explaining the benefits of cross-

platform applications using .NET Core, carried in the January 2019 issue of OSFY (<https://bit.ly/3oA6hlO>).

If you have any classic .NET Framework applications, it's time to consider migrating to .NET 5, which offers cross-platform support and performance improvements. It's a great time to consider .NET 5 and EF 5 Core for developing new apps as they support cloud native features out-of-the-box. This includes support for gRPC, Project Tyre (which is a .NET microservices and orchestration

tool), and REST APIs, as well as faster cross-platform development with Windows Subsystem for Linux (WSL) and Linux.

Why should I care about Entity Framework Core?

As discussed, EF Core can serve as an object-relational mapper (ORM), which helps .NET developers to work with a database using .NET objects. It also eliminates the need for most of the data-access code that typically needs to be coded.

1. EF Core works with many databases, including SQL (on-premises and Azure), SQLite, MySQL, PostgreSQL and Azure Cosmos DB. In simple terms, it's a data access API that supports non-relational databases like Azure Cosmos DB too.
2. It supports 23+ databases, including prominent ones like Oracle, MySQL, MariaDB, PostgreSQL, FireBird, DB2, Informix, Progress Open Edge, etc. You can check <http://aka.ms/efcoreproviders> for more information on the databases it supports.
3. We can use EF Core to build various types of .NET projects, say, Blazor app, WPF, Winforms, ASP.NET Core and Xamarin, across the spectrum.

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.NET 5 Release Candidate

Single file applications
Smaller container images
Web and cloud investments
Windows ARM64 support
Continued performance improvements
New C# 9.0, F# 5.0 language features

dotnet/get-dotnet5

Figure 1: Top features of .NET 5